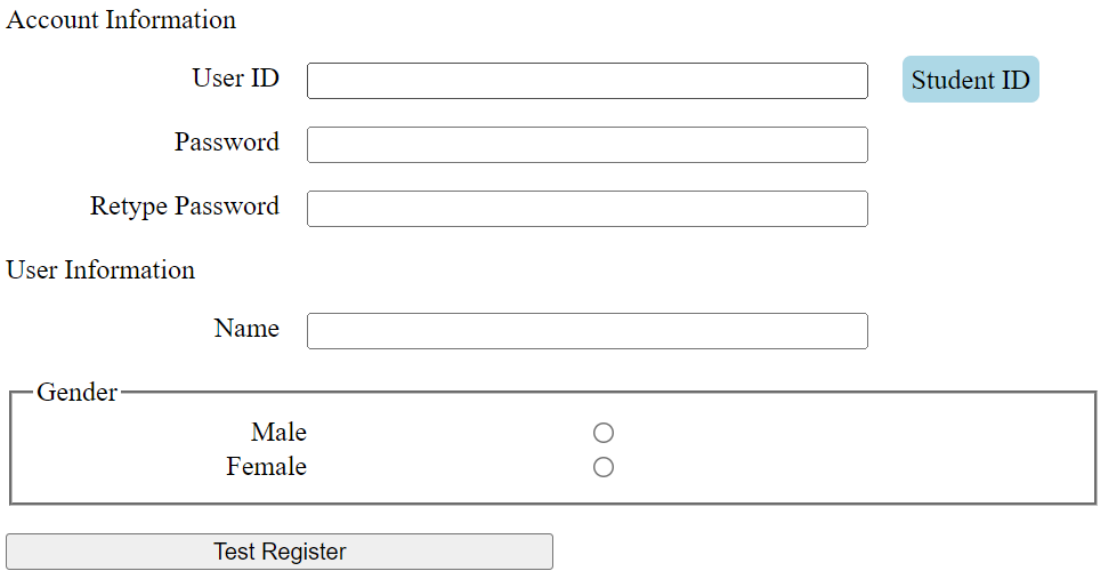
Lab Exercises 7 – JavaScript Part 1

Aims

- To practice how to create interactivity for HTML pages using JavaScript and CSS, while maintaining clear separation of HTML, CSS, and JS file.
- To review JavaScript variables and expression.
- To gain the skills and knowledge to complete Assignment 2.

Task 1: Create Interactivity using JavaScript

Dynamic behaviour can be added to presentation of a Web page using CSS e.g., with pseudo classes like a:hover or input:focus, or through JavaScript. In this lab we will demonstrate the use of simple pseudo classes and JavaScript to enhance the user interactions, by displaying a 'tooltip' when the mouse pointer moves over an <input> element. Remember to always design the form carefully before you start coding. Below is the expect output from this lab exercise.



Step 1: HTML Creation

- Use notepad++ to open and edit regform2.html. Review the HTML and locate the 'comments' where will add missing code.
- Format the regform2.html code based on the instructions below:

```
<!DOCTYPE HTML>
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="description" content="Web development" />
    <meta name="keywords"    content="Registration Form" />
    <meta name="author"      content="put your name here" />
    2) Link to desktop.css
    3) Link to tooltip.css
    4) Link to tooltip.js
    <title>Web Development Registration Form</title>
</head>
<body>
    <form method="post" action="abc@gmail.com">
        <p>Account Information</p>
        <p>
            <label for="sid" >User ID</label>
            <input type="text" name="sid" id="sid" />
                1) Write tooltip HTML
        </p>
        <p>
            <label for="pwd1">Password</label>
            <input type="password" name="pwd1" id="pwd1" />
```

1) Link desktop.css to regform2.html using <link>. Certain attributes are needed for <link> to work properly.
2) Link tooltip.css to regform2.html using <link>. Certain attributes are needed for <link> to work properly.
3) Link desktop JavaScript to regform2.html using <script></script>.
4) Create the tooltip using :

   Student ID

   Note: the span element is used, as the content is not a paragraph or another division. Span is an inline element and thus will not occupy the full form width.

Step 2: CSS For the Form

- Use notepad++ to open and edit desktop.css. Review the CSS and locate the 'comments' where will add missing code.
    - Form
        - Set the width to 40em;
    - Label
        - Float to left;
        - Align text to right;
        - Set width to 10em;
        - Set right margin to 1em;
    - Input
        - Set width to 50%;
    - Set .textinput
        - Set margin to 10px;
    - Set .radioinput fieldset
        - Remove borders;
    - Set .radioinput legend
        - Float to left;
        - Align text to right;
        - Set width to 9em;
    - Set .radioinput input
        - Set width to 2em;
    - Set .radioinput label
        - Disable float;
        - Align text to left;
    - Set .buttoninput
        - Align text to right;
    - Set .buttoninput input
        - Set width to auto;

Step 3: CSS for the Tooltip

- Use notepad++ to open and edit tooltip.css. Review the CSS and locate the 'comments' where will add missing code.
    - tooltip

```
display      : _______________;  /* tool tips are initially not visible */
position     : _______________;  /* place tooltip according to the order in the HTML */
_______      : lightblue;        /* uses a lightblue background */
border-radius: 5px;
_________    : center;           /* center text inside the tooltip */
padding      :                   /* add 5px spacing around the text */
left         : 1 em;
```

- tooltip:after
  position : absolute;
  _____ : _____; /* set the top to 10px */
  Border-style : solid;
  _____ : _____; /* set the height to 0px */
  _____ : _____; /* set the width to 0px */

Step 4: JavaScript Creation (for the Tooltip)

- Use notepad++ to open and edit tooltip.js. Create and complete function showTip().

```
function showTip () {
    var sidTip = document.getElementById("sidTip");
    //obtain access to the HTML tool tip element
    //using id "sidTip" and link it a varialbe named sidTip

    sidTip.style.display = "inline";
    //display the HTML tool tip element sidTip bying change
    //the value of its CSS property "display" to "inline"
}
```

- Create function hideTip(), using showTip() as an example.

```
function hideTip () {  //this function hides the tool tip

//first, get access to the tool tip HTML element

//then, hide the tool tip HTML element by changing its CSS property "display" to "none"
}
```

- Create function init(). In function init(), we link JavaScript to the appropriate events of corresponding HTML elements.

```
function init () { /* the init function links functions to appropriate
                      events of corresponding HTML elements */

/* create a variable named sid.get access to the HTML element by its id
"sid" and link it to sid */

 /* link function showTip to the onmouseover event of sid */
 /* link function hideTip to the onmouseout event of sid  */
}

/* link function init to the onload event of the window so
that function init will be called when the page is loaded */
```

Step 5: Validate the page and fix any errors displayed and revalidate.

- Use the validator at http://validator.w3.org to validate your html file.
- Use the CSS validator at http://jigsaw.w3.org/css-validator/ .